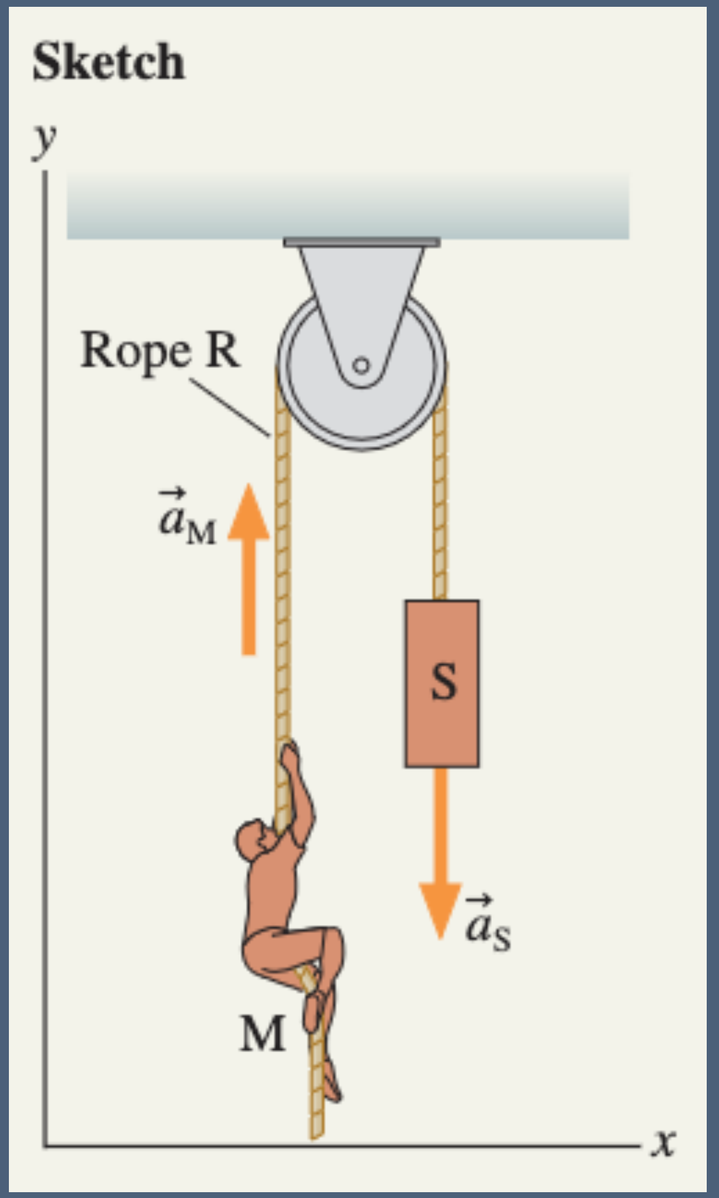
Problem Solving Session

10/13/2026

Example

- A $200\,\mathrm{kg}$ set used in a play is stored in the loft above the stage. The rope holding the set passes up an over a pulley, then is tied backstage. The director tells a $100\,\mathrm{kg}$ stagehand to lower the set. When he untiles the rope, the set falls and the unfortunate man is hoisted into the loft. What is the stagehand's acceleration?

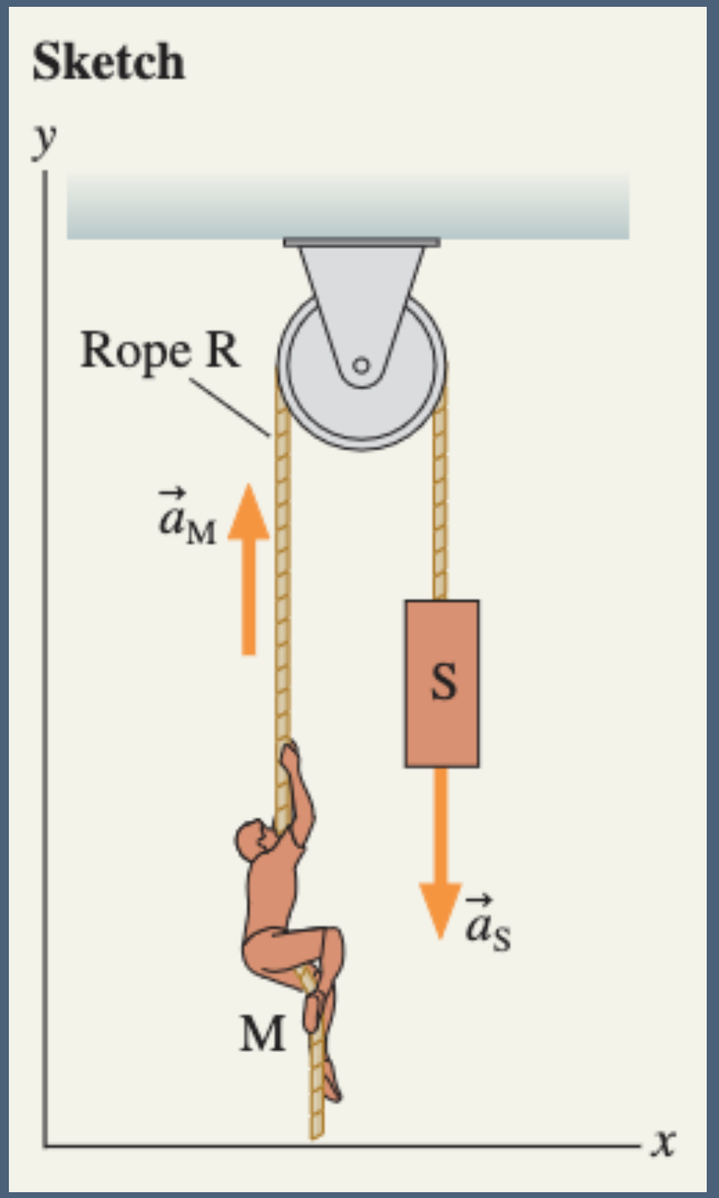


Example

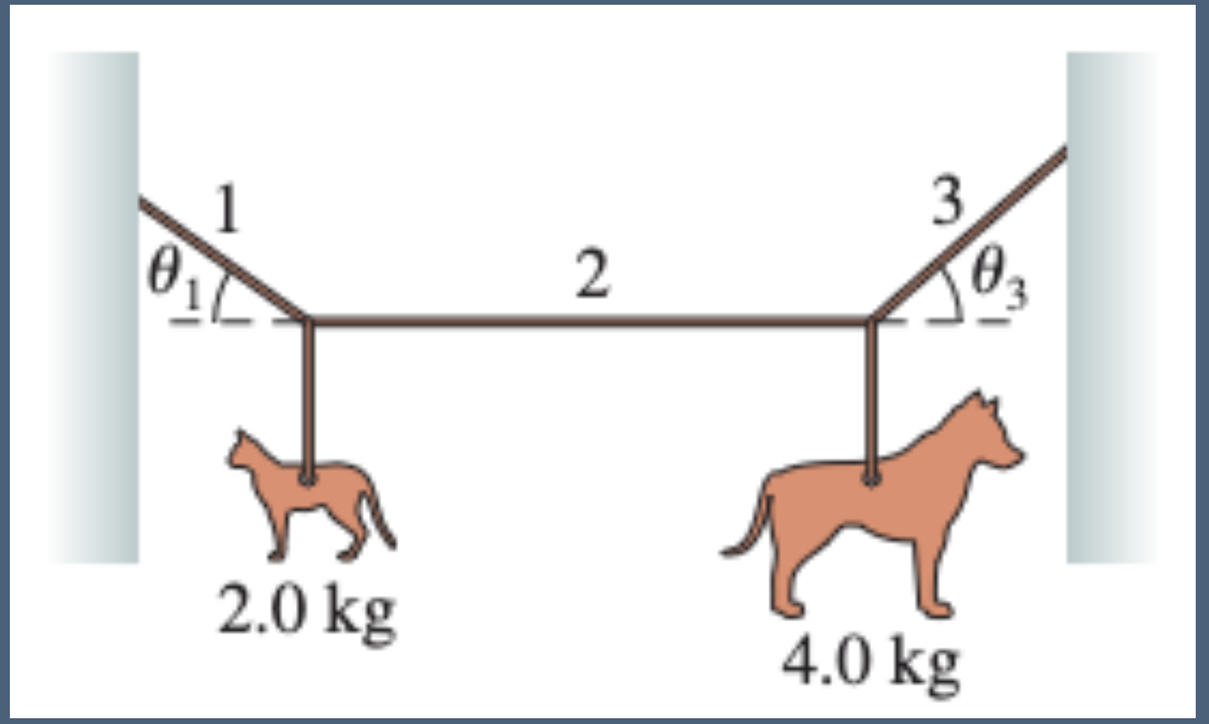
- Because to objects are connected they have the same acceleration (acceleration constraint)

  $a_{yS} = a_{yP} = a_y$

- $a = g\dfrac{m_S - m_P}{m_S + m_P} = 3.27\ \mathrm{m/s^2}$



- A mobile at the art museum has a $2.0\,\mathrm{kg}$ steel cat and a $4.0\,\mathrm{kg}$ steel dog suspended from a lightweight cable, as show in the image. It is found that $\theta_1 = 20°$ when the center cope is adjusted to be perfectly horizontal. What are the tension and the angle of rope 3?



- The century-old ascensores in Valparaiso, Chile, are picturesque cable cares built on stilts to keep the passenger compartments level as they go up and down the steep hillsides. As shown in the figure, one care ascends as the other descends. The cars use a two-cable arrangement to compensate for friction; one cable passing around a large pulley connects the cars, the second is pulled by a small motor. Suppose the mass of each car (with passengers) is $1500\,\mathrm{kg}$, the coefficient of rolling friction is 0.020 and the cars move at constant speed. What is the tension in (a) the connecting cable and (b) the cable to the motor?

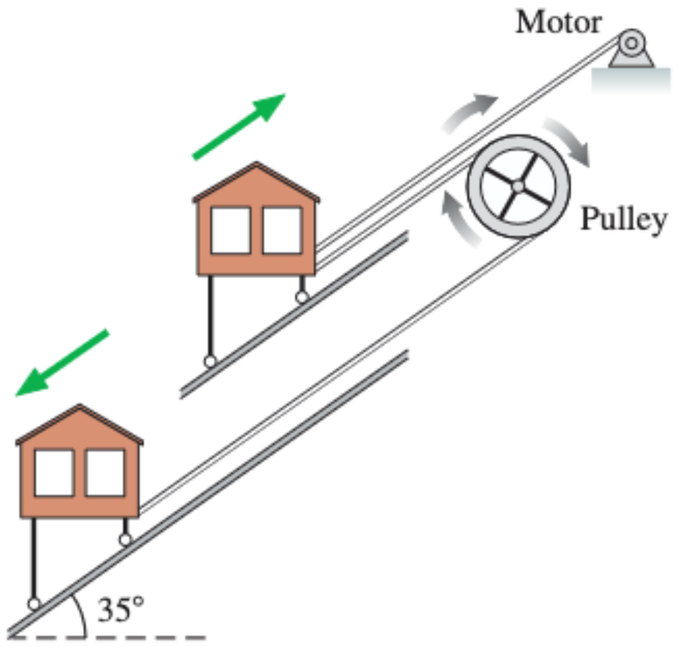


FIGURE P7.43